# EDUCONNECT+

(Complete University Website + CMS)

Software Requirements Specification

Release 1.0

18 Jun, 2024

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# Revision History

| Revision | Revision Date | Description | Author/Modifier |
|---|---|---|---|
| v1.0 | 18 Jun, 2024 | Initial Document | Musharaf Ali/ Usama Riaz |
| v1.1 | 28 July, 2024 | Second Document | Musharaf Ali/Usama Riaz/ Nasir Mehmood |
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# Document Approval

The following Software Requirements Specification has been accepted and approved by the following:

| Signature | Printed Name | Title | Date |
|---|---|---|---|
| | Dr. Ghulam Ali | Lead Software Eng. | |
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# Table of Contents

# 1. INTRODUCTION

EduConnect+ is a revolutionary web-based University Management System meticulously crafted to streamline academic administration, foster effective communication, and facilitate performance tracking within educational institutions. This comprehensive software solution seamlessly integrates a user-friendly website and a dynamic dashboard, catering to the diverse needs of users across different roles and responsibilities.

## 1.1 PURPOSE

The purpose of this section is to introduce EduConnect+, a sophisticated web-based University Management System, engineered to harmonize various facets of academic administration, communication, and performance tracking within educational institutions. This section provides an overview of the primary goals and intentions behind the development of EduConnect+.

## 1.2 SCOPE

EduConnect+ is designed to address the challenges faced by educational institutions in managing administrative tasks, fostering communication, and effectively tracking academic performance. The scope of EduConnect+ encompasses a user-friendly website and a dynamic dashboard, catering to users with diverse roles and responsibilities within the academic setting.

## 1.3 DEFINITIONS, ACRONYMS, AND ABBREVIATIONS

To ensure clarity and a shared understanding among stakeholders, this section provides precise definitions for terms, acronyms, and abbreviations used throughout the document. It serves as a reference guide to promote consistency in communication about EduConnect+.

## 1.4 REFERENCES

The references section lists all documents relevant to the understanding and development of EduConnect+. These may include academic papers, technical specifications, or external resources that contribute to the comprehensive understanding of the project.

**Development Resource:**

- Bootstrap
- Tailwind Css
- React Icons
- MongoDB Altas
- ChatBot

**SRS Reference**:

- http://regisindia.com/wp-content/uploads/2013/10/SRS-Development-of-Web-GIS-Tool_IEEE-SA-830-SRS-Format_Final_27082013.pdf
- https://en.wikipedia.org/wiki/Software_requirements_specification
- http://www.cse.chalmers.se/~feldt/courses/reqeng/examples/srs_example_2010_group2.pdf

## 1.5 OVERVIEW

EduConnect+ integrates a web-based University Management System, blending a user-friendly website and a dynamic dashboard. The system addresses various aspects:

**EduConnect+ Website:**
The website serves as a gateway to an array of features, enhancing the overall user experience. Key pages include:

- **HomePage**: The central hub offering a quick overview of essential updates, notifications, and access points to various sections of the system.
- **AboutPage**: A detailed section providing insights into the EduConnect+ platform, its objectives, and the benefits it brings to users.
- **Contact Us**: A dedicated channel for users to communicate with administrators, fostering effective collaboration and issue resolution.
- **FacultyPage**: An informative section designed for faculty members, offering resources, announcements, and tools to support their academic responsibilities.
- **Admission Page**: A streamlined process for managing and processing student admissions efficiently.
- **Events**: An interactive calendar showcasing upcoming events, seminars, and academic activities within the institution.

**EduConnect+ Dashboard:**
The dashboard acts as a powerful tool, centralizing and simplifying University management tasks. It focuses on three main pillars:

- **University Management**: Streamlining administrative processes, creating and organizing classes, and managing the overall structure of the educational institution.
- **Class Organization**: Facilitating smooth class organization, allowing administrators to create, modify, and monitor classes efficiently.
- **Student and Faculty Management**: Providing a centralized platform for adding, updating, and managing student and faculty profiles, ensuring accurate and up-to-date information.

**Key Functionalities:**
EduConnect+ is engineered to enhance the efficiency and effectiveness of academic processes by offering key functionalities:

- **Attendance Tracking**: Seamlessly monitoring and recording student attendance for each class, simplifying the task for faculty members.

- **Performance Assessment**: Empowering teachers to assess student performance by providing marks and constructive feedback, fostering a culture of continuous improvement.
- **Record Access and Visibility**: Granting users the ability to access academic records, view marks, and track performance, promoting transparency and accountability.
- **Effortless Communication**: Facilitating communication between students, faculty, and administrators within the system, ensuring quick and effective collaboration.

In essence, EduConnect+ emerges as a comprehensive solution, designed to meet the diverse needs of academic institutions. It provides a user-friendly interface, robust functionality, and a centralized platform for efficient University management.

# 2. General Description

The General Description of EduConnect+ provides an overarching perspective on the software's purpose, functionality, and intended users. It outlines EduConnect+'s role as a comprehensive web-based University Management System, designed to streamline administrative tasks, enhance communication, and facilitate performance tracking within educational institutions. This section sets the stage for a deeper exploration of the software's features, constraints, and dependencies, offering stakeholders a clear understanding of its significance and potential impact within the academic landscape.

## 2.1 Product Perspective

EduConnect+ is positioned as a pivotal tool within educational institutions, functioning as a centralized platform to streamline administrative tasks, enhance communication, and facilitate performance tracking. It stands as an integral component within the ecosystem of educational management systems, providing seamless integration with existing infrastructure while offering a comprehensive suite of features tailored to meet the diverse needs of users.

## 2.2 Product Functions

EduConnect+ encompasses a wide array of functions designed to address the multifaceted requirements of academic administration, communication, and performance tracking. These functions include but are not limited to:

- **Admin Dashboard:** A dynamic interface tailored for administrators to efficiently manage various aspects of academic operations, including class organization, student enrollment, and data management.
- **Attendance Tracking:** Automated systems for monitoring and recording student attendance, providing real-time insights to faculty members and administrators.
- **Performance Assessment:** Tools to assess student performance, assign grades, and provide feedback, fostering a culture of academic excellence and continuous improvement.
- **Data Visualization:** Visual representations of academic data, facilitating informed decision-making processes for administrators and educators.

- **Communication:** Integrated communication channels allowing seamless interaction between students, faculty, and administrators, promoting collaboration and transparency within the educational community.

## 2.3 User Characteristics

EduConnect+ caters to a diverse user base within educational institutions, each with distinct roles, responsibilities, and requirements:

- **Administrators:** Individuals responsible for overseeing the overall functioning of the educational institution, including managing resources, setting policies, and ensuring compliance with regulatory standards.
- **Faculty Members:** Educators tasked with delivering academic content, assessing student performance, and providing guidance and support to students.
- **Students:** Learners engaged in academic pursuits, seeking access to resources, guidance from faculty, and avenues for academic and personal development.

## 2.4 General Constraints

While EduConnect+ aims to offer a comprehensive solution for educational management, it operates within certain constraints, including:

- **Technological Limitations:** Dependency on existing technological infrastructure and compatibility with hardware and software systems prevalent within educational institutions.
- **Regulatory Compliance:** Adherence to regulatory frameworks governing data privacy, security, and accessibility within the educational sector.
- **Resource Constraints:** Limitations in terms of budgetary allocations, human resources, and timeframes for development and implementation.

## 2.5 Assumptions and Dependencies

The development and successful deployment of EduConnect+ rely on certain assumptions and dependencies, including:

- **Availability of Resources:** Assumption of access to necessary human, financial, and technological resources required for development and implementation.
- **Stakeholder Cooperation:** Dependence on the cooperation and engagement of stakeholders, including administrators, faculty members, students, and IT personnel.
- **Technology Partnerships:** Collaboration with technology partners or vendors for access to specialized tools, platforms, or services integral to the functioning of EduConnect+.

# 3. Specific Requirements

Specific Requirements outline the detailed functionalities and characteristics that EduConnect+ must possess to fulfill its objectives effectively. These requirements serve as the foundation for the software's development and implementation, ensuring that it meets the specific needs of users and adheres to industry standards and best practices. From external interface specifications to functional features like attendance tracking and performance assessment, each requirement is carefully crafted to contribute to the overall functionality, usability, and reliability of the system. By clearly defining these requirements, stakeholders can align expectations, guide development efforts, and ultimately deliver a robust and user-friendly University Management System that meets the demands of modern educational institutions.

## 3.1 External Interface Requirements

### 3.1.1 User Interfaces

- The user interface of EduConnect+ shall be intuitive, user-friendly, and accessible across various devices and screen sizes.
- It shall feature a responsive design to ensure optimal usability on desktops, laptops, tablets, and mobile devices.
- The interface shall incorporate modern design principles, including clear navigation, consistent layout, and intuitive controls, to enhance user experience.
- Accessibility features such as keyboard navigation, screen reader compatibility, and high contrast options shall be implemented to accommodate users with disabilities.
- The user interface shall provide customization options, allowing users to personalize their experience based on preferences such as language, theme, and layout.

### 3.1.2 Hardware Interfaces

- EduConnect+ shall be compatible with commonly used hardware devices including desktop computers, laptops, tablets, and smartphones.
- It shall not impose specific hardware requirements beyond those typical of modern computing devices, ensuring broad compatibility and accessibility.

### 3.1.3 Software Interfaces

- EduConnect+ shall integrate with standard web browsers such as Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge.
- It shall leverage web technologies including HTML, CSS, and JavaScript for front-end development, ensuring cross-platform compatibility and performance.
- Backend services shall be implemented using appropriate programming languages and frameworks, facilitating seamless communication between the user interface and underlying data systems.

### 3.1.4 Communications Interfaces

- EduConnect+ shall support various communication protocols and APIs for data exchange between different modules and external systems.
- It shall integrate with email services to enable automated notifications, reminders, and communication between users within the system.
- Real-time communication channels such as chat or messaging features may be incorporated to facilitate instant interaction between users, administrators, and support staff.

## 3.2 Functional Requirements

### 3.2.1 Admin Dashboard

- The Admin Dashboard shall provide administrators with a centralized interface for managing various aspects of academic operations.
- It shall allow administrators to create, modify, and delete user accounts, including students, faculty members, and staff.
- The dashboard shall feature tools for class organization, enabling administrators to create, schedule, and manage classes, as well as assign teachers and students to respective classes.
- It shall provide access to student enrollment and admission functionalities, allowing administrators to process admissions, manage student profiles, and track enrollment statistics.
- The dashboard shall offer data management capabilities, including the ability to generate reports, analyze trends, and export data for further analysis or integration with external systems.

### 3.2.2 Attendance Tracking

- The Attendance Tracking module shall enable faculty members to record student attendance for each class session.
- It shall provide options for manual entry or automated tracking, allowing faculty members to choose the most suitable method based on their preferences and requirements.
- The module shall support real-time attendance monitoring, providing instant feedback to faculty members and administrators regarding student attendance status.
- It shall offer reporting and analytics features, allowing administrators to track attendance trends, identify patterns, and take necessary actions to improve attendance rates.

### 3.2.3 Performance Assessment

- The Performance Assessment module shall facilitate the assessment of student performance through various evaluation methods such as quizzes, assignments, and exams.
- It shall allow teachers to assign grades, provide feedback, and track student progress over time.
- The module shall support customizable grading criteria and rubrics, enabling teachers to evaluate student performance based on predefined standards and objectives.
- It shall offer features for generating performance reports, analyzing student performance data, and identifying areas for improvement.

### 3.2.4 Data Visualization

- The Data Visualization module shall provide visual representations of academic data, including attendance records, grades, and performance metrics.
- It shall offer interactive charts, graphs, and dashboards to present data in a meaningful and easily understandable format.
- The module shall support customization options, allowing users to customize the visualization settings, filters, and parameters according to their preferences.
- It shall facilitate data-driven decision-making by providing insights into trends, patterns, and correlations within the academic data.

### 3.2.5 Communication

- The Communication module shall enable seamless communication between students, faculty members, administrators, and other stakeholders within the system.
- It shall support various communication channels such as messaging, email, and announcements, allowing users to choose the most appropriate channel for their communication needs.
- The module shall provide features for group communication, enabling users to collaborate on projects, discuss assignments, and share resources.
- It shall offer notification capabilities, allowing users to receive timely updates, reminders, and alerts regarding important events, deadlines, and announcements.

## 3.3 Use Cases

### 3.3.1 Admin adds new student

- The use case involves the admin adding a new student to the system.
- **Preconditions**: Admin must be logged in to the system.
- **Basic Flow:**
    1. Admin navigates to the student management section of the admin dashboard.
    2. Admin selects the option to add a new student.
    3. Admin fills in the required student details such as name, contact information, and enrollment details.
    4. Admin saves the information, and the system generates a unique student ID.
    5. The new student is added to the system database.
- **Alternate Flow:**
    - If any required information is missing or invalid, the system prompts the admin to correct the errors before saving the information.

### 3.3.2 Admin creates class

- The use case involves the admin creating a new class within the system.
- **Preconditions**: Admin must be logged in to the system.
- **Basic Flow**:
    1. Admin navigates to the class management section of the admin dashboard.
    2. Admin selects the option to create a new class.
    3. Admin enters the class details such as course name, class code, and schedule.
    4. Admin assigns a teacher to the class and adds enrolled students.
    5. Admin saves the information, and the new class is added to the system.
- **Alternate Flow:**
    - If any required information is missing or invalid, the system prompts the admin to correct the errors before saving the information.

### 3.3.3 Teacher Takes Attendance

- The use case involves a teacher taking attendance for a class session.
- **Preconditions**: Teacher must be logged in to the system and assigned as the teacher for the specific class.
- **Basic Flow:**
    1. Teacher navigates to the class dashboard or attendance section.
    2. Teacher selects the class session for which attendance needs to be taken.
    3. Teacher views the list of enrolled students for the selected class session.
    4. Teacher marks each student as present, absent, or tardy.
    5. Teacher saves the attendance information, and it is recorded in the system.
- **Alternate Flow:**
    - If any issues arise during the attendance-taking process, such as network connectivity problems, the system provides an option to retry or save the information later.

### 3.3.4 Teacher Assesses Performance

- The use case involves a teacher assessing the performance of students for a specific assignment, quiz, or exam.
- **Preconditions**: Teacher must be logged in to the system and have access to the grading module.
- **Basic Flow:**
    1. Teacher navigates to the grading section or selects the specific assignment/quiz/exam to assess.
    2. Teacher views the list of students who submitted the assignment or completed the assessment.
    3. Teacher reviews each student's submission, evaluates their performance, and assigns grades or scores.
    4. Teacher provides feedback and comments on each student's performance.
    5. Teacher saves the grading information, and it is recorded in the system.
- **Alternate Flow:**
    - If any technical issues occur during the grading process, such as system errors or data loss, the system provides an option to save progress and resume later.

### 3.3.5 Student Views Marks

- The use case involves a student viewing their grades and marks for various assignments, quizzes, and exams.
- **Preconditions**: Student must be logged in to the system and enrolled in the respective classes.
- **Basic Flow:**
    1. Student navigates to the grades or marks section of the student dashboard.
    2. Student selects the class for which they want to view grades.
    3. Student views a list of assignments, quizzes, or exams along with corresponding grades or marks.
    4. Student can click on individual assignments to view detailed feedback and comments provided by the teacher.
    5. Student can download or print the grade report for their records.
- **Alternate Flow:**
    - If the grade report is not available or there are technical issues accessing the grades, the system provides an option to contact the teacher or administrator for assistance.

## 3.4 Non-Functional Requirements

### 3.4.1 Performance

- The system shall respond to user interactions within an acceptable timeframe, with page load times not exceeding 3 seconds under normal operating conditions.
- It shall support concurrent user sessions without significant degradation in performance, capable of handling a minimum of 1000 simultaneous users.

### 3.4.2 Reliability

- The system shall operate reliably with a minimum uptime of 99.9% over a specified period, ensuring minimal downtime for maintenance or unforeseen issues.
- It shall have mechanisms in place to handle errors gracefully, providing informative error messages and ensuring data integrity in case of system failures.

### 3.4.3 Availability

- The system shall be available 24/7, with scheduled maintenance windows communicated to users in advance.
- It shall have redundant server infrastructure and failover mechanisms to minimize downtime and ensure continuous availability of services.

### 3.4.4 Security

- The system shall adhere to industry-standard security practices, including encryption of sensitive data, secure authentication mechanisms, and role-based access control.
- It shall implement measures to protect against common security threats such as SQL injection, cross-site scripting (XSS), and unauthorized access attempts.

### 3.4.5 Maintainability

- The system shall be designed with modularity and scalability in mind, allowing for easy maintenance and future enhancements.
- It shall provide comprehensive documentation for administrators, developers, and end-users, covering installation procedures, system architecture, and usage guidelines.

### 3.4.6 Portability

- The system shall be compatible with modern web browsers including Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge.
- It shall support multiple operating systems such as Windows, macOS, and Linux, ensuring broad accessibility across different computing platforms.

## 3.5 Inverse Requirements

Inverse requirements outline conditions or actions that the system should avoid or prevent. These requirements focus on undesirable outcomes or behaviors that must be minimized or eliminated to ensure the effectiveness and usability of the system.

## 3.6 Design Constraints

Design constraints specify limitations or restrictions imposed on the design and implementation of the system. These constraints may arise from factors such as technology limitations, budgetary constraints, regulatory requirements, or organizational policies. Adhering to design constraints is essential for ensuring that the system meets the specified objectives while complying with relevant standards and guidelines.

## 3.7 Logical Database Requirements

Logical database requirements define the structure, organization, and relationships of data within the system's database. These requirements detail how data should be stored, accessed, and manipulated to support the system's functionality and user requirements. They include specifications for data entities, attributes, relationships, constraints, and integrity rules, ensuring that the database meets the needs of the system and its users.

## 3.8 Other Requirements

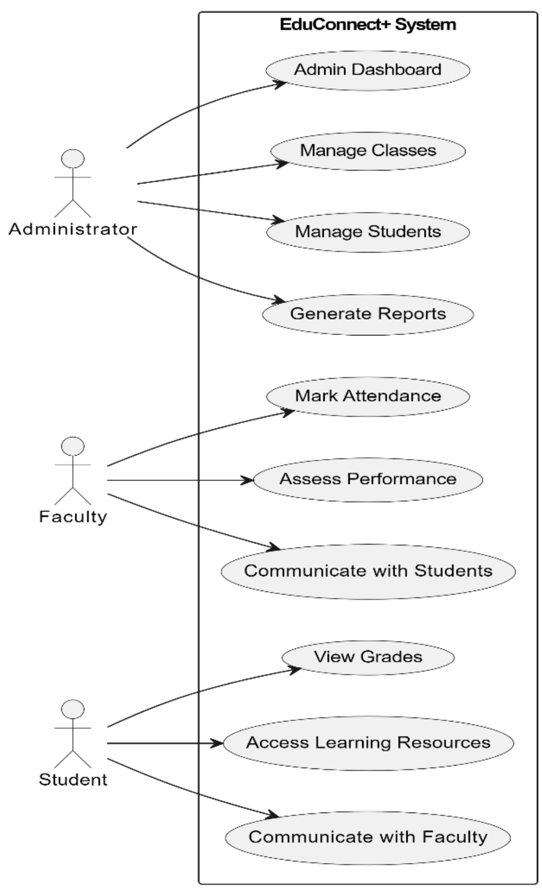
Other requirements encompass any additional specifications or criteria that are not covered by the preceding sections but are essential for the successful development and implementation of the system. These may include legal or regulatory requirements, performance benchmarks, quality standards, or specific user preferences or expectations. Addressing these requirements ensures that the system meets all relevant criteria and delivers value to its users.

# 4. Analysis Models

Analysis models provide visual representations and descriptions of the system's behavior, structure, and interactions. They help stakeholders understand and analyze the system's requirements, functionality, and processes.

## 4.1 Use Case Diagrams

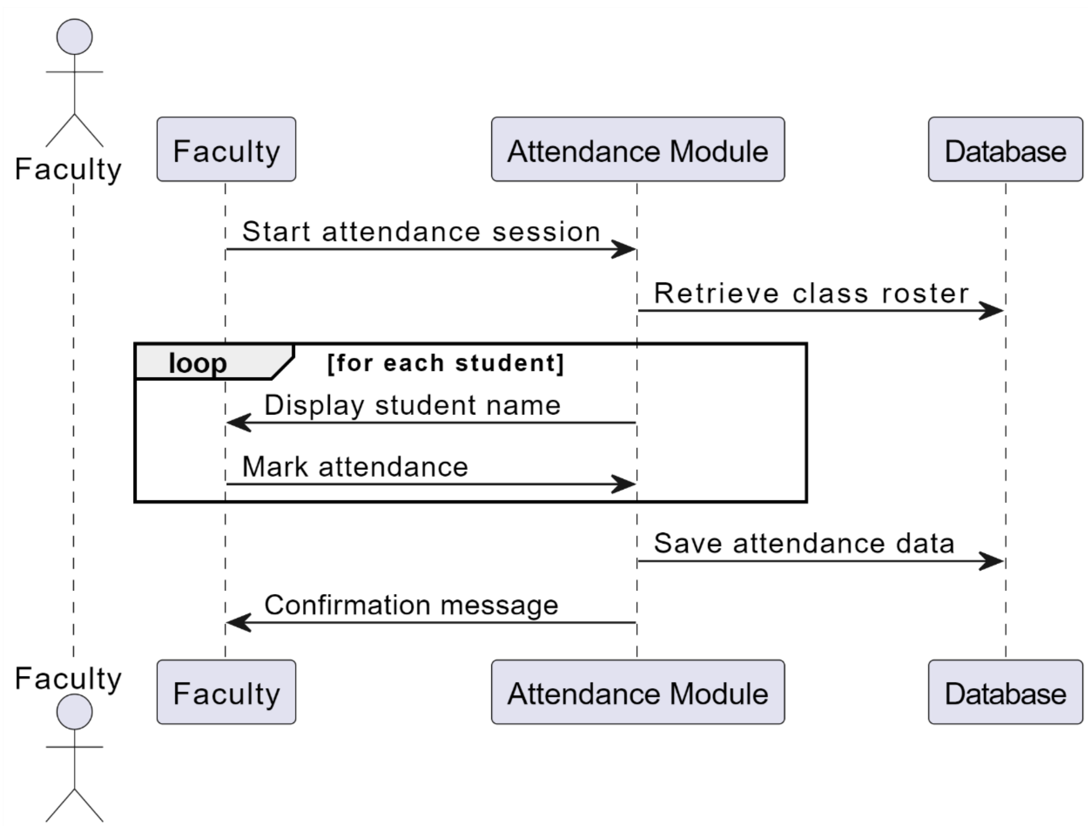
This figure provides a visual representation of the use cases within the EduConnect+ system. It illustrates the interactions between actors and system components, showcasing the functionalities available to different user roles.



## 4.2 Sequence Diagrams

Sequence diagrams depict the interactions between different components or actors within the system over time. They illustrate the flow of messages or actions between objects or entities, showing the sequence of events and the order in which they occur. Sequence diagrams are useful for modeling the dynamic behavior of the system and identifying dependencies and interactions between system elements.

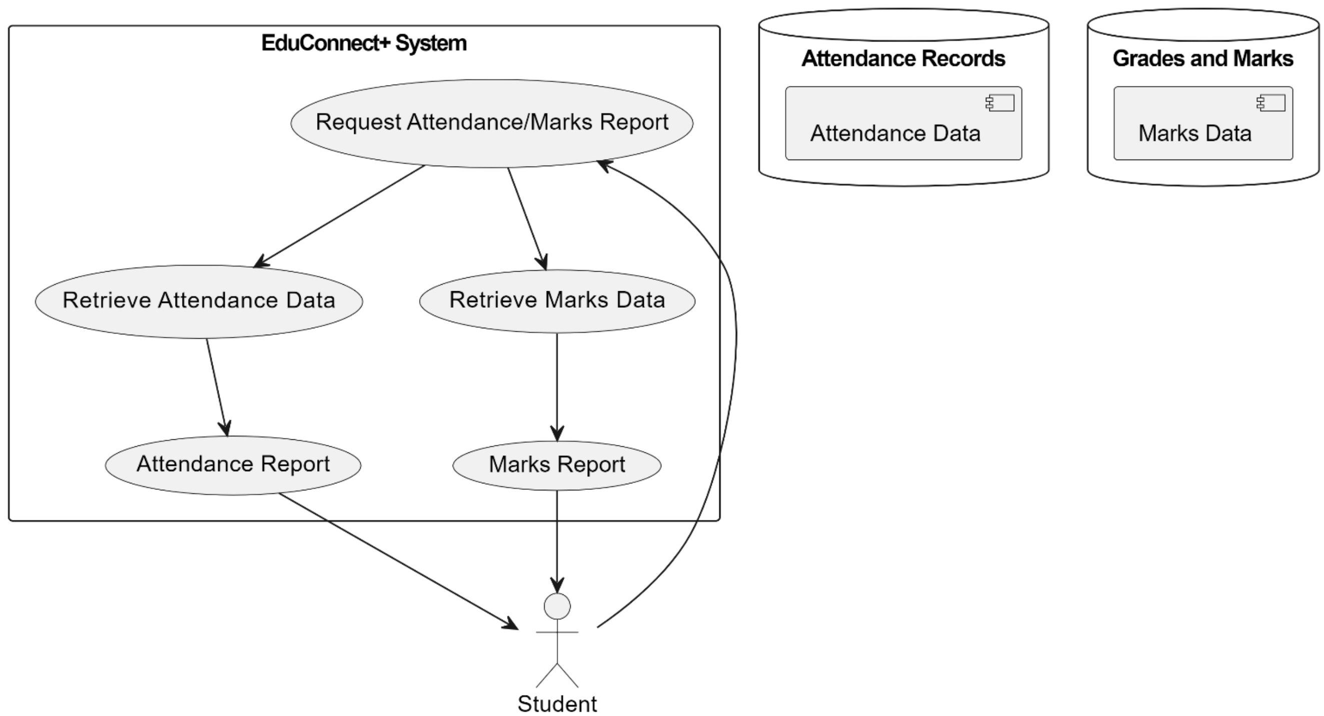
**Sequence Diagram (Faculty Marks Attendance):**




## 4.3 Data Flow Diagrams (DFD)

Data Flow Diagrams (DFD) represent the flow of data within the system, showing how information moves between processes, data stores, and external entities. They provide a high-level view of the system's data processing and transformation processes, highlighting the inputs, outputs, and data flows between different components. DFDs are useful for understanding the data requirements of the system and identifying potential bottlenecks or inefficiencies in data processing.

**Data Flow Diagram** illustrating the flow of data for student attendance and marks report within the EduConnect+ system:



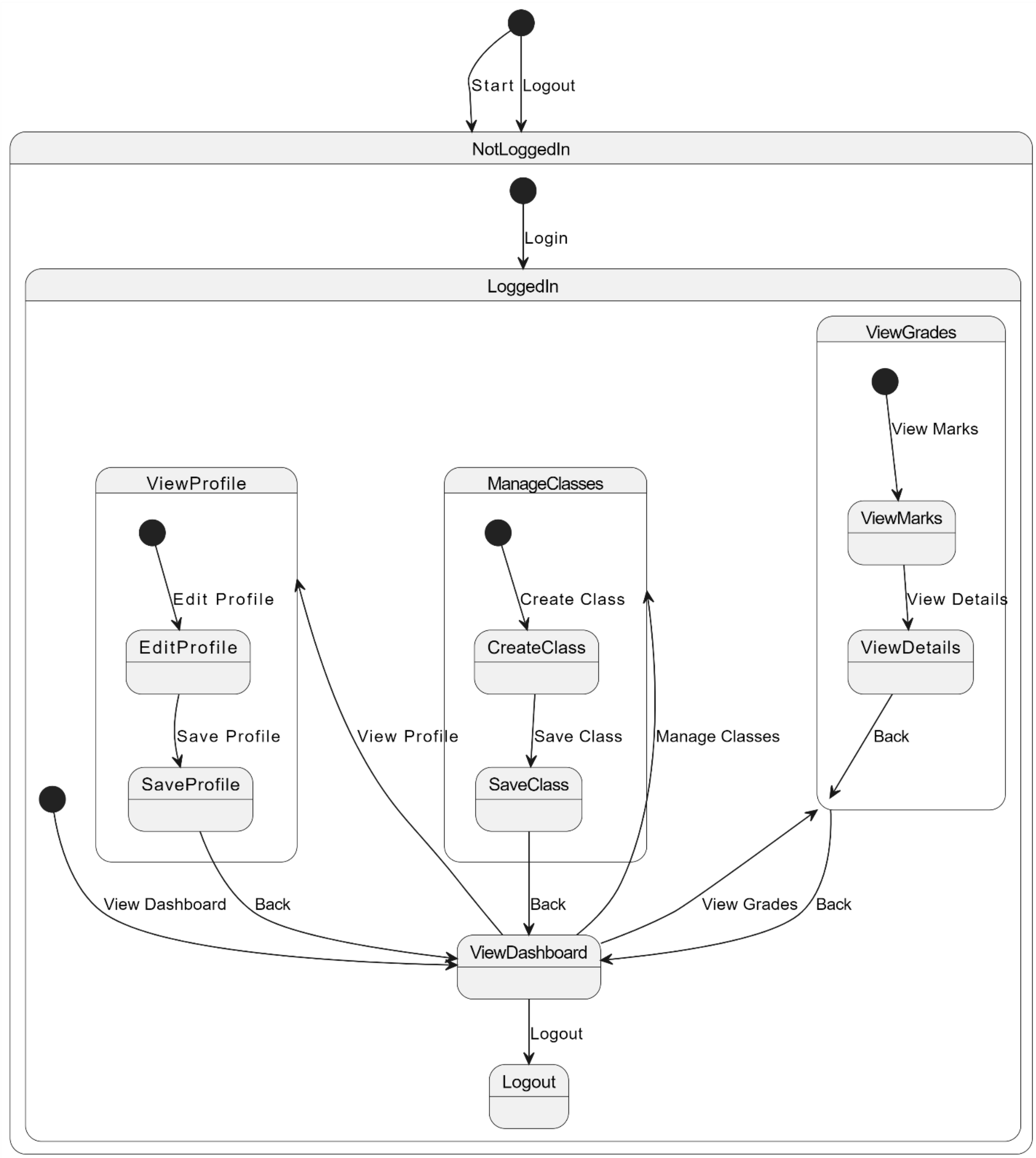

## 4.4 State-Transition Diagrams (STD)

State-Transition Diagrams (STD) model the behavior of a system or component as a finite set of states and the transitions between these states triggered by events or actions. They describe the lifecycle of objects or entities within the system, showing how they transition from one state to another in response to external stimuli. STDs are particularly useful for modeling systems with complex behavior or multiple states, such as user interfaces, control systems, or communication protocols.

**In this diagram:**

- The user starts in the "**NotLoggedIn**" state.
- Upon logging in, the user transitions to the "**LoggedIn**" state and can access the dashboard.
- From the dashboard, the user can navigate to view their profile, manage classes, view grades, or log out.
- The user can edit their profile, create classes, view marks, or view grade details from their respective states.
- Finally, the user can log out from any state to return to the "**NotLoggedIn**" state.

**State-Transition Diagram** (STD) illustrating the lifecycle of a user account within the EduConnect+ system:



## 5. Change Management Process

Change management is a systematic approach to managing changes to a system, process, or organization in a controlled and structured manner. It ensures that changes are effectively planned, implemented, monitored, and evaluated to minimize disruptions and maximize benefits. The change management process typically includes the following stages:

1. **Identification of Change:**
    - Identification of the need for change, whether initiated internally or externally, based on factors such as technological advancements, business requirements, regulatory changes, or stakeholder feedback.
2. **Change Request:**
    - Submission of a formal change request outlining the proposed change, including its rationale, scope, objectives, and potential impacts.
3. **Change Evaluation:**
    - Evaluation of the change request to assess its feasibility, potential risks, benefits, and alignment with organizational goals and priorities.
4. **Change Approval:**
    - Review and approval of the change request by designated stakeholders, such as project sponsors, steering committees, or change control boards, based on established criteria and guidelines.
5. **Change Planning:**
    - Development of a detailed change management plan outlining the steps, resources, responsibilities, and timelines required to implement the change effectively.
6. **Change Implementation:**
    - Execution of the change plan, including communication of the change to relevant stakeholders, training and support for affected individuals, and deployment of necessary resources.
7. **Change Monitoring:**
    - Ongoing monitoring and tracking of the change implementation process to ensure adherence to the plan, identify any issues or deviations, and take corrective actions as needed.
8. **Change Review:**
    - Evaluation of the outcomes and impacts of the change implementation, including its effectiveness in achieving the intended objectives, and identification of lessons learned and areas for improvement.

9. **Change Documentation:**
    - Documentation of all aspects of the change management process, including change requests, approvals, plans, implementation activities, monitoring reports, and review outcomes, for future reference and audit purposes.

10. **Continuous Improvement:**
    - Incorporation of feedback and insights gained from the change management process into ongoing improvement efforts, to enhance organizational resilience, agility, and adaptability in response to evolving challenges and opportunities.

# 6. Appendices

The appendices section provides supplementary information, references, or detailed data that supports the content presented in the main document. It includes additional resources, documents, or materials that may be relevant to stakeholders seeking further clarification or insight into specific aspects of the project.

## A.1 Appendix 1

Appendix 1 contains detailed technical specifications, diagrams, or charts that complement the main document. It may include:

- Technical diagrams such as network diagrams, system architecture diagrams, or entity-relationship diagrams.
- Detailed tables or charts presenting additional data or statistics related to the project.
- Code snippets, algorithms, or programming examples referenced in the main document.
- Any additional information or documentation deemed relevant to the understanding or implementation of the project.

## A.2 Appendix 2

Appendix 2 provides supplementary documentation, references, or resources that expand upon topics discussed in the main document. It may include:

- References to academic papers, research studies, or industry reports cited in the main document.
- Additional reading materials, websites, or online resources for further exploration of related topics.
- Glossary of terms, acronyms, or abbreviations used throughout the document to aid readers in understanding technical or specialized terminology.
- Any other relevant information or materials that contribute to a comprehensive understanding of the project or its context.